

# Dr. Aniket Chitre

E-mail: [ac2349@cantab.ac.uk](mailto:ac2349@cantab.ac.uk) | Mobile: +44 7943180767 | LinkedIn: <https://www.linkedin.com/in/aniket-chitre/>

## Relevant Experience

### Unilever

**Digital R&D Manager, Unilever Fragrance House, London UK** Nov 2025 – Present

- Project manager for a major external partnership developing applied AI tools to accelerate fragrance creation and improve product performance.
- Scientific lead on an external R&D collaboration applying computational chemistry and ML methods to a core technical perfumery problem.
- Deliver internal data science solutions for analytical datasets and lead development of Python-based tools for perfumers.
- Define the UFH modelling roadmap and chair an internal modelling community of practice.

**Modelling & Automation Manager, Materials Innovation Factory, Liverpool UK** July – Oct 2025

- Trained ML (tabular foundation) models to guide next-gen shampoo chassis development.
- Coordinated a team of three lab scientists to run the automated formulation programme: planned and prioritised global studies, managed execution and data delivery to stakeholders.
- Provided thought leadership to define the R&D strategy for digital and AI-driven formulation.

**Senior Research Scientist, Science & Technology, Port Sunlight UK** Aug 2024 – June 2025

- Developed internal ML solutions, e.g., deployed our first ML-driven process optimiser via AzureML, which demonstrated 20% consumer property improvement for a hair care product.
- Drove lab automation by leading internal R&D projects and collaborating with external automation vendors; led validation of a £1M high-throughput platform and liquid/solids dosing.
- Supervised postgraduate researchers for Unilever's AI and automation collaborations.

**PhD Researcher, BASF** 2020 – 2024

**University of Cambridge, UK (2 years)**

**Cambridge CARES and A\*STAR, Singapore (2 years)**

- Developed ML/AI and lab automation methods suitable for accelerating liquid formulation design. Thesis available at: <https://www.repository.cam.ac.uk/handle/1810/376242>.
- Established an end-to-end process of 'making and measuring' formulation samples driven by a machine learning-guided workflow, including characterisation via computer vision.
- Published five peer-reviewed articles, all in Q1 journals (Nature Scientific Data, Computational Chemical Engineering, Digital Discovery), and presented in four international conferences.
  - Developed an ML-guided design of experiments method for screening novel formulations.
  - Published an 800+ formulation sample dataset, doubling previous efforts in the literature.
  - Built a self-driven pH adjustment robot with a chemistry-informed ML algorithm.

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## Education

**PhD in Chemical Engineering, Hughes Hall, University of Cambridge** 2020 – 2024

- Thesis: "Accelerating Liquid Formulation Design using **Lab Automation and Machine Learning**".
- Supervisors: Prof. Alexei Lapkin (Cambridge), Prof. Kedar Hippalgaonkar (Singapore).

**M.Eng. in Chemical Engineering (77%, Dean's List), Imperial College London** 2016 – 2020

- Core chemical engineering classes (mathematics, chemistry, transport phenomena, reaction engineering, safety and loss prevention, environmental engineering, process control etc.)
- Specialised in process systems engineering (optimisation, dynamical systems etc.) and took the following business electives: economics, project management, managing innovation.
- Key projects included final year design, techno-economic, and master's research project.

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## **Highlighted Scholarships and Awards**

- BASF, PhD sponsorship 2020 – 2024: all tuition fees, research costs and stipend covered (£150k).
  - Henry Royce Institute, 2024: Scholarship for programme at Cambridge Judge Business School.
  - National University of Singapore, 2022: Best Oral Presentation at international conference.
  - Imperial College London, 2016 – 2020: Dean's List (top 10% of students) in all years of studies.
  - Massachusetts Institute of Technology, 2019: Fully funded summer research exchange (£6k).
  - BP Award, Imperial College, 2017: Awarded to top performing student with significant extra-curricular contributions - £1.5k cash prize.
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## **Key Skills**

### **Programming & Machine Learning**

- Proficient in Python (6+ years); working knowledge of R and MATLAB. Experienced with standard libraries including NumPy, pandas, matplotlib, scikit-learn, TensorFlow etc. Use Git for version control and collaborative workflows. Comfortable with agentic AI coding tools, e.g., Claude code.
- Applied ML/AI methods to experiment design/optimisation and molecular modelling. Developed supervised and unsupervised models, ML-guided workflows, and domain-specific algorithms.
- Completed the 1-year *Machine Learning for Science* programme delivered by Cambridge Spark.

### **Project & People Management**

- Associate Project Management Diploma (2025–2026), including e-learning, 1:1 coaching, and on-the-job application on R&D projects.
- Managed Year-in-Industry students at Unilever and supervised undergraduate researchers during PhD at A\*STAR, Singapore.

### **Cross-Functional & Interdisciplinary Collaboration**

- Led a multi-disciplinary team of engineers, scientists, and statisticians to deliver PhD project goals; collaborated across experimental and computational domains.

### **Innovation & Entrepreneurship**

- Completed *EnterpriseTECH*, a 10-week deep-tech innovation programme at Cambridge Judge Business School, Spring 2024. Consulted for a med-tech start-up on market validation and commercialisation strategy.
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## **Selected Publications**

Full scholarly profile available at: <https://scholar.google.com/citations?user=DeIMN4oAAAAJ&hl=en>.

- Chitre, A., Woods, D. & Lapkin, A. A. Machine Learning-Guided Space-filling Designs for High Throughput Liquid Formulations Development. *Comp. Chem. Eng.* **195**, 109007 (2025).
  - Chitre, A. *et al.* Accelerating Formulation Design via Machine Learning: Generating a High-throughput Shampoo Formulations Dataset. *Sci Data* **11**, 1–10 (2024).
  - Chitre, A. *et al.* pHbot: Self-Driven Robot for pH Adjustment of Viscous Formulations via Physics-informed-ML. *Chemistry Methods* **4**, e202300043 (2023).
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## **Interests**

- **Sport** – Captained my college badminton team at the University of Cambridge, leading them to runners-up out of 29 colleges. Played hockey at university level, rowed for Hughes Hall, and played table tennis competitively for Imperial College London. Continue to play badminton regularly and enjoy watching football and cricket.
- **Travel & international experience** – Enjoy exploring new places through travel and work. Have lived in the US briefly and in Singapore (including relocating during the pandemic), which broadened my global perspective and adaptability in unfamiliar environments.